

PROBLEM STATEMENT

Power Rule & Quotient Rule Combination

Find the derivative with respect to x of the function:

$$y = \left(\frac{x^2 + 1}{x^2 - 1} \right)^3$$

STEP 1 প্রদত্ত মূলদ ঘাত ফাংশনটি উল্লেখ করি (Given Rational Function)

$$y = \left(\frac{x^2 + 1}{x^2 - 1} \right)^3$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} \left[\left(\frac{x^2 + 1}{x^2 - 1} \right)^3 \right]$$

STEP 3 শক্তি বিধি ও চেইন রুল প্রয়োগ করি ($\frac{d}{dx}[u^3] = 3u^2 \frac{du}{dx}$)

$$\frac{dy}{dx} = 3 \left(\frac{x^2 + 1}{x^2 - 1} \right)^2 \cdot \frac{d}{dx} \left(\frac{x^2 + 1}{x^2 - 1} \right)$$

STEP 4 ভাগবিধি প্রয়োগ করে অভ্যন্তরীণ ভগ্নাংশের অন্তরক নির্ণয় করি (Applying Quotient Rule)

$$\frac{d}{dx} \left(\frac{x^2 + 1}{x^2 - 1} \right) = \frac{(x^2 - 1)(2x) - (x^2 + 1)(2x)}{(x^2 - 1)^2} = \frac{2x(x^2 - 1 - x^2 - 1)}{(x^2 - 1)^2} = \frac{-4x}{(x^2 - 1)^2}$$

STEP 5 মানগুলো গুণ করে হর ও লবকে সাজিয়ে চূড়ান্ত অন্তরক পাই (Final Algebraic Simplification)

$$\frac{dy}{dx} = 3 \cdot \frac{(x^2 + 1)^2}{(x^2 - 1)^2} \cdot \left[\frac{-4x}{(x^2 - 1)^2} \right] = -\frac{12x(x^2 + 1)^2}{(x^2 - 1)^4}$$

FINAL ANSWER
$$\frac{dy}{dx} = -\frac{12x(x^2 + 1)^2}{(x^2 - 1)^4}$$

BOX 1: FORMULATION & COMBINED POWER-QUOTIENT RULE

- **Quotient Derivative:** For $\frac{u}{v}$, $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$.
- For $u = x^2 + 1$, $v = x^2 - 1$:

$$vu' - uv' = (x^2 - 1)(2x) - (x^2 + 1)(2x) = 2x(-2) = -4x$$

- Combining with outer power: $\frac{dy}{dx} = 3 \cdot \frac{u^2}{v^2} \cdot \frac{-4x}{v^2} = -\frac{12xu^2}{v^4}$.

BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Quotient Rule Sign Error:** Flipping terms to $(uv' - vu')$ which erroneously flips the final sign from negative to positive.
- **Denominator Exponent Error:** Multiplying $(x^2 - 1)^2 \times (x^2 - 1)^2$ as $(x^2 - 1)^4$. A common mistake is writing $(x^2 - 1)^3$.

BOX 3: REAL DOMAIN & SINGULARITIES

- **Domain of Definition:** The rational expression requires $x^2 - 1 \neq 0 \iff x \neq \pm 1$.

$$\text{Domain} = \mathbb{R} \setminus \{-1, 1\}$$

- **Symmetry & Critical Point:** $y(x)$ is an even function ($y(-x) = y(x)$). Its derivative $\frac{dy}{dx}$ is an odd function.
- At $x = 0$, $\frac{dy}{dx} = 0$ and $y(0) = (-1)^3 = -1$ (local maximum on $(-1, 1)$).

BOX 4: LOGARITHMIC DIFFERENTIATION ALTERNATIVE

- Take natural logarithm: $\ln y = 3 [\ln(x^2 + 1) - \ln(x^2 - 1)]$.
- Differentiating: $\frac{1}{y} \frac{dy}{dx} = 3 \left[\frac{2x}{x^2+1} - \frac{2x}{x^2-1} \right] = 6x \left[\frac{(x^2-1)-(x^2+1)}{x^4-1} \right] = -\frac{12x}{x^4-1}$.
- Multiplying by $y = \frac{(x^2+1)^3}{(x^2-1)^3}$:

$$\frac{dy}{dx} = -\frac{12x}{(x^2+1)(x^2-1)} \cdot \frac{(x^2+1)^3}{(x^2-1)^3} = -\frac{12x(x^2+1)^2}{(x^2-1)^4}$$

BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 4 / 5 MARKS)

- **1.0 Mark:** Correct power rule setup $3(\dots)^2 \cdot \frac{d}{dx}(\dots)$.
- **2.0 Marks:** Accurate algebraic execution of Quotient Rule yielding $\frac{-4x}{(x^2-1)^2}$.
- **1.0 Mark:** Correct exponents combination and final boxed fraction $-\frac{12x(x^2+1)^2}{(x^2-1)^4}$.